

Green Hydrogen and the Logic of Green Industrial Policy: A Global Assessment of Instruments and Implementation; Insights from machine learning

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Abstract

As countries race to decarbonise their economies, green hydrogen has emerged as a critical technology for reducing emissions in hard-to-abate sectors. This thesis investigates whether green industrial policy (GIP) offers a coherent and effective framework to support the development of green hydrogen, with a focus on the instruments, implementation patterns, and international dynamics shaping current strategies. Drawing on a self-constructed dataset of 107 policy interventions across 33 countries since September 2023, the study employs hierarchical agglomerative clustering (i.e., machine learning technique) to identify common patterns in the design and orientation of green hydrogen policies. The results highlight a strong global bias toward supply-side instruments—primarily grants and tax credits—with limited demand-side support mechanisms. This imbalance raises concerns about the long-term viability of market creation for green hydrogen. The thesis also discusses the risks of incoherence and lack of coordination, particularly in developing countries, and argues that effective industrial policy must be both targeted and internationally compatible to avoid a race to the bottom. Finally, the thesis reflects on data limitations, the difficulty of comparing policies across jurisdictions, and the potential for future research using supervised machine learning techniques. Overall, the findings suggest that while green industrial policy is necessary, it must evolve toward greater standardisation, transparency, and demand-side support to fully unlock the potential of green hydrogen¹

¹All the figures shown in this thesis are created by the author, with the exception of the illustrative image in Box 5. [Click here to access the repository with the relevant code and sources.](#)

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1 Introduction

The global energy sector is undergoing a profound transformation as efforts to tackle climate change and strengthen energy security accelerate the move toward cleaner, more sustainable sources. Interest in low-emissions hydrogen has surged because it is widely recognised as a key solution for decarbonising hard-to-abate sectors such as heavy industry, shipping and aviation. The recent energy crisis has highlighted its ability to enhance energy security and encouraged governments to weave it into their net-zero strategies; and new industrial policies in several major economies now place hydrogen technologies at their core (IEA, 2024). Yet, despite this momentum, the widespread adoption of low-emissions hydrogen has not taken off, and significant challenges remain before its full potential can be realised.

While there are numerous approaches to producing low-emissions hydrogen, one of the most promising is ‘green’ (renewable) hydrogen, which is generated by electrolysis powered by renewable energy sources. At scale, the transition to green hydrogen can contribute to global net-zero targets in four important ways: First, hydrogen is a promising substitute for fossil fuels in the energy sector—where its use is expected to grow markedly by 2050—and in industrial processes that require high temperatures; the only by-product of hydrogen combustion is water, so it could improve the sustainability of hard-to-abate sectors where electrification is technically challenging. Second, hydrogen is an essential input for developing fuel-cell-based vehicles, which are likely to play a critical role in reducing emissions in logistics, the built environment, and sectors such as mining. Third, hydrogen is a key feedstock for several chemical industries, so renewable hydrogen could significantly support the decarbonisation of carbon-intensive products such as fertilisers while also enabling the emergence of new synthetic fuels. Lastly, as an energy-storage medium, hydrogen can help integrate electricity produced from intermittent renewable sources (Cammeraat, Dechezleprêtre and Lalanne, 2022).

Box 1. The different shades of hydrogen

Many international organisations, national administrations and private-sector actors use a “colour” code to indicate the production method—and therefore the typical carbon intensity—of hydrogen. An overview of these classifications is as follows:

- **Green (renewable) hydrogen** – produced by electrolysis (splitting water into hydrogen and oxygen) powered by renewable electricity. When burned, it is effectively carbon-neutral because the only by-product is water; when used as a feedstock, it enables low- or zero-carbon products and hydrogen-derived fuels.
- **Blue hydrogen** – produced from natural gas (or coal) via steam-methane reforming or coal gasification, with a significant share of the resulting CO₂ captured and stored in deep geological formations (carbon capture, utilisation, and storage, CCUS). Critics argue that blue hydrogen investment could slow the shift to full decarbonisation by competing with green hydrogen, but proponents see it as a short-term step for decarbonising existing grey-hydrogen facilities and helping the market to mature.
- **Grey hydrogen** – also produced from natural gas via steam-methane reforming, but

without CCUS, so the CO₂ is released to the atmosphere.

- **Purple hydrogen** – produced using nuclear energy.
- **Brown/black hydrogen** – produced from coal.

There is growing criticism of the colour code itself. Using colours—or unqualified terms such as “sustainable,” “low-carbon” or “clean”—can mask differences in actual emissions, and the lack of standardised definitions complicates contracts and investment decisions. Nevertheless, while recognising that there presently lacks an international agreement on the use of different terminologies for hydrogen production methods, this thesis uses ‘green hydrogen’ to denote hydrogen produced using electrolysis from renewable energy sources and ‘low-emissions hydrogen’ to denote hydrogen produced using different combinations of renewable technologies and carbon mitigation technologies.

Sources: Adapted from Australian Government, 2024 and OECD, 2025.

At present, over 99 % of global hydrogen is produced using unabated fossil fuels. This largely reflects the technological maturity—and therefore lower cost—of fossil-fuel-based production, in contrast to the still-nascent and comparatively expensive green-hydrogen technologies. Consequently, most national strategies aim to narrow the cost gap between green hydrogen and its fossil-fuel-derived equivalents, thereby fostering green-hydrogen markets and improving its commercial viability as a low-carbon fuel and feedstock (OECD 2025; IEA 2024).

In 2023, global hydrogen production reached 97 million tonnes—an increase of 2.5 % over 2022. According to the International Energy Agency (IEA), meeting future demand will require output to rise to about 500 million tonnes by 2050, with more than 90 % expected to come from renewable sources (IEA 2024; IEA 2022). Without government support, producing green hydrogen is estimated to cost roughly USD 5.3 per kilogram in the European Union and USD 5.7 per kilogram in the United States, compared with around USD 1.9 and USD 1.3, respectively, for grey hydrogen (Menon Economics 2023). This gap is driven mainly by the price of electrolyzers and the higher cost of renewable electricity relative to natural gas. It has recently widened further because inflation and falling fossil-fuel prices have lowered the cost of grey hydrogen (IEA 2024). High production costs therefore continue to constrain the uptake of low-carbon hydrogen under current market conditions.

Many governments have responded with comprehensive strategies to bridge this gap. Their plans typically combine measures to expand supply, stimulate demand, build transport and storage infrastructure, regulate use and quality, and facilitate foreign trade. The IEA (2024) estimates that nearly USD 100 billion in public funds was announced, enacted or allocated to hydrogen projects in 2023 alone. Support instruments range from tax incentives and grants to soft loans, land concessions, and tradable energy certificates. But why is state support so crucial? Criscuolo *et al.* (2022) argue that coordinating the low-carbon transition is inherently complex, requiring governments to take an increasingly interventionist, steering role—often manifested in a renewed interest in industrial policy—both to drive systemic change at home and to adapt to it internationally.

Moreover, how effective are today’s policies for green hydrogen, and should they prioritise the

supply side or the demand side? Current policy packages for green hydrogen remain tilted toward the supply side, and their effectiveness is therefore uneven. The International Energy Agency (IEA) calculates that public funding for production incentives is roughly one-and-a-half times larger than the support available for stimulating demand (IEA, 2024). In practical terms, the measures announced so far would create demand for only about 10.5 million tonnes of low-emission hydrogen per year by 2030, whereas official production targets imply an output of 35 to 43 million tonnes (IEA, 2024). Closing this gap would generate substantial economic and climate benefits because robust demand-side instruments—such as long-term offtake agreements, quotas in hard-to-abate sectors, carbon contracts for difference, or green-procurement mandates—can signal a clear economic rationale for redirecting research and development toward low-carbon technologies even before a fully developed market exists. By reducing policy and market uncertainty, such instruments help firms justify investment in higher-risk but more sustainable options, hasten large-scale deployment, and accelerate cost reductions through learning effects and economies of scale (World Bank, 2023).

Furthermore, should emerging and developing economies embark on large-scale green-industrial policies for hydrogen? Most developing countries are currently absent from the core sectors of renewable energy and green hydrogen, which raises the question of whether green industrial policy is even relevant for them. However, the real challenge lies in determining how these countries should respond to the current wave of green industrial policy being implemented by advanced economies. This is particularly complex given that multilateral institutions have long advised developing countries to rely on market mechanisms and avoid active industrial policy (Reed, 2024). In this new global context—where major economies are heavily subsidising green technologies—the traditional advice appears increasingly outdated, and developing countries must now reconsider their strategic approach.

2 Literature Review

2.1 The global landscape of hydrogen and green hydrogen’s potential

Hydrogen production continues to grow steadily. In 2023, global hydrogen supply rose by 2.5 % compared to 2022, reaching around 97 million tonnes (Mt). Most of the demand was met through domestic production within industrial hubs, with limited international trade so far. China leads in production, making up about 30 % of the global total, followed by the United States and the Middle East at 14 % each, and India at 9 % (IEA, 2024). A major driver of optimism for green hydrogen is the declining cost of renewable energy, especially solar and wind. As electricity is the main input in electrolytic hydrogen production, cheaper power makes it more cost-competitive. Electrolysers themselves are also becoming more efficient and scalable, further improving the economics of production (IEA, 2023).

A key advantage of green hydrogen lies in its ability to decarbonize hard-to-abate sectors (industries where direct electrification is technically difficult or prohibitively expensive). These include sectors like steelmaking, cement, chemicals, shipping, and long-haul aviation. For example, in the steel industry, green hydrogen can replace coal-based blast furnaces through direct reduced iron (DRI) processes, drastically lowering emissions. Similarly, in the chemicals sector, hydrogen is essential for producing ammonia and methanol, which are used both as fuels and industrial feedstocks. In transport, hydrogen-based fuels (like ammonia or synthetic kerosene) are emerging as

alternatives for maritime shipping and aviation, where batteries are too heavy or have insufficient energy density (IEA, 2021, IRENA, 2022). Green hydrogen also plays a strategic role in energy system flexibility. It can serve as a long-term storage solution for intermittent renewables, helping to balance supply and demand on the grid. In this capacity, it supports grid reliability and enables sector coupling, where hydrogen links the electricity system to hard-to-abate sectors like steel, cement, heavy-duty transport, and chemicals (IRENA, 2022).

Beyond its decarbonisation potential, hydrogen contributes to several broader policy goals. It enhances energy security by enabling storage of surplus renewable electricity, which can be later transformed into hydrogen or hydrogen-derived fuels like ammonia and synthetic methane. And unlike fossil fuels, which depend on continuous imports that can be disrupted for political or strategic reasons, renewable electricity sources rely on locally available inputs—sunlight and wind—which cannot be “cut off” by a foreign supplier. Green hydrogen also improves air quality when used in fuel cells or blended into gas grids, reducing pollution in urban areas (IEA, 2019). Economically, hydrogen offers countries an opportunity to revitalize industrial sectors and create jobs. This is especially relevant for fossil fuel-dependent regions where existing infrastructure—such as pipelines and refineries—can be adapted for hydrogen use, helping to smooth the transition toward cleaner energy systems (IEA, 2022).

Despite this promise, several challenges persist. The sector faces high capital requirements, regulatory uncertainty, and underdeveloped infrastructure for transport, storage, and cross-border trade. These hurdles contribute to systemic risk, making private investment difficult without policy support. As argued by Goldberg et al. (2024), sectors with high learning spillovers and capital intensity—like semiconductors and hydrogen—require targeted public intervention to reach scale, since early-stage investments benefit from both technology maturation and demand creation (Goldberg et al., 2024, NBER). If industrial policy can successfully reduce investment barriers and coordinate market development, green hydrogen could become a cornerstone of the clean energy transition. It offers both climate and industrial benefits, especially for countries rich in renewable resources, like Chile, Morocco, and Mongolia, which are well-positioned to become competitive exporters (Cammeraat et al., 2022).

2.2 The economic rationale behind Green Industrial Policy GIP

The need for green industrial policy (GIP) in the case of green hydrogen comes from two main market failures: a *static* externality in the form of environmental damage not priced into markets, and a *dynamic* externality from learning-by-doing, which causes firms to underinvest in innovation and R&D due to knowledge spillovers (*see table 1 below*). Traditionally, economists have been skeptical of industrial policy, often citing concerns about government inefficiency, rent-seeking, and political capture. As Krugman (1994) warned, industrial policy has historically suffered from being co-opted by interest groups, while Rodrik (2004) and Chang (2002) have highlighted the informational problem faced by governments in identifying the right sectors or technologies to support. In the international context, there is also the fear of triggering a “race to the bottom”, where countries retaliate by subsidising competing industries, leading to global resource misallocation rather than productivity gains. Nonetheless, the application of industrial policy in areas such as climate change, renewables, and electric vehicles is increasingly viewed as justified—if not essential—due to structural market failures in these sectors. In particular, nascent clean technologies like green hydrogen

are subject to positive production externalities (learning-by-doing, technology spillovers). However, these features also create opportunities for economies of scale, dynamic comparative advantage and first-mover advantages, which private markets alone may not exploit without public support.

Moreover, the green transition requires proactive support due to the urgency and global nature of climate crisis. Traditional market-based tools like carbon pricing often fall short. As Rodrik (2014) argues, green technologies face a “double market failure”: they are simultaneously undervalued due to the underpricing of carbon and underinvested in due to weak private R&D incentives. Similarly, Hallegatte et al. (2013) explain that even with correct carbon pricing, private investment may not occur fast enough. Green technologies like green hydrogen face high upfront costs and uncertain payoffs. Because of path dependence, existing fossil-based systems get “locked in”, which discourages clean innovation. According to Aghion et al. (2019), this lock-in requires public policy to “tilt” markets towards cleaner paths. Without this intervention, it becomes harder and more expensive to switch to low-carbon technologies later. This logic is also supported by Acemoglu et al. (2012), who show that early and temporary interventions (e.g., carbon taxes and R&D subsidies) can redirect innovation before fossil-intensive industries become too dominant. Delaying policy action increases transition costs. Therefore, GIP is not just useful—it is necessary for shifting the economy towards clean technologies like green hydrogen.

Category	Rationale	Description	Sources
Market Failures	Externalities (static)	Costs or benefits not reflected in market prices (e.g. emissions).	Rodrik (2014); Acemoglu et al. (2012)
	Externalities (dynamic)	Learning-by-doing: innovation benefits spill over to others.	Aghion et al. (2019); Rodrik (2014)
	Coordination Failures	Private actors fail to invest without complementary infrastructure or skills.	Rodrik (2004); Chang (2002)
	Increasing Returns to Scale	Larger production lowers unit costs; early support helps reach scale.	Rodrik (2014); Krugman (1994)
	Infant Industry Argument	New industries need protection until they become competitive.	Chang (2002); Rodrik (2014)
	Information Asymmetry	Government can help correct cases where firms lack full market info.	Rodrik (2014); Aghion et al. (2019)
Geostrategic Goals	Terms-of-Trade Improvement	Enhance national control over valuable inputs or exports.	Rodrik (2004); Mazzucato (2013)
	National Security	Secure critical industries or technologies from foreign dependency.	Rodrik (2022); Goldberg et al. (2024)
	Chokepoint Protection	Defend against strategic vulnerabilities in supply chains.	Goldberg et al. (2024); IEA (2024)
	Green Transition / Climate Change	Accelerate decarbonisation via public support for clean tech.	Rodrik (2014); Acemoglu et al. (2012)
	Resilience	Build capacity to respond to shocks (e.g. pandemics, war, climate).	Hallegatte et al. (2013); Rodrik (2022)

Table 1: Economic rationales for industrial policy.

2.3 What does GIP look like?

Green industrial policy employs a spectrum of instruments that support environmentally beneficial technologies. According to Hallegatte et al., 2013, these instruments can be grouped as: First,

“sunrise” measures—subsidies, tax incentives and concessional finance—help early-stage markets such as green hydrogen to scale; learning-by-doing and knowledge spill-overs make such support efficient (Hallegatte et al., 2013). Second, “sunset” measures—temporary assistance for sectors adversely affected by decarbonisation, such as coal mining—ease the social and political costs of transition by averting sudden job losses. Third, innovation policy—public investment in R&D, university research and state-led development—remains vital; Mazzucato’s “The Entrepreneurial State” shows that many breakthrough technologies—from biotech to IT—benefited from early public risk-taking, and green hydrogen will likely need the same. Finally, regulations and performance standards—renewable-energy mandates or emissions limits, for instance—push firms toward cleaner alternatives even where explicit carbon pricing is absent. Under good institutions, these tools reshape production structures and foster an innovation-friendly environment for clean technologies.

Objective	Typical tools	Key references
Stimulate new green industries (<i>sunrise</i>)	R&D grants, production subsidies, concessional loans, public procurement	Rodrik 2014; Hallegatte <i>et al.</i> 2013
Ease decline of polluting sectors (<i>sunset</i>)	Time-bound subsidies, retraining support, regional adjustment funds	Hallegatte <i>et al.</i> 2013
Create markets	Renewable-energy mandates, efficiency standards, guaranteed offtake contracts	Aghion <i>et al.</i> 2019
Reduce investment risk	State development banks, loan guarantees, co-investment funds	Mazzucato 2013

Table 2: Green-Industrial-policy objectives, typical policy tools.

2.4 Effectiveness and political feasibility

The success of GIP depends on its design and political context. Rodrik (2014) highlights two main criticisms: (1) that governments can’t “pick winners,” due to the informational problem they face (*see sections 2.2* above) and (2) that such policies lead to rent-seeking. He argues both can be addressed with good institutional design. A well-functioning GIP should allow for errors and correction, not perfection. Mistakes are part of the learning process, and too few mistakes may actually mean too little experimentation. In practice, policies aimed at creating “green jobs” or national competitiveness are more politically appealing than abstract climate goals. But this sometimes leads to misallocation—like investing in solar panels not where it’s most efficient, but where it creates local jobs. Still, Rodrik suggests this might be better than nothing. He also argues that a global “green subsidy race” is more productive than a trade war, as it expands clean technology supply. Hallegatte et al. (2013) also warn about “capture” by vested interests. Their guidelines for success include transparency, measurable goals, and regular evaluation. Past examples from Asia show industrial policy can succeed if it adapts to changing conditions.

2.5 Lessons from GIP: The case of Jordan

Jordan presents an interesting case of energy policy reform without a full green transition. As a country that imports nearly all of its natural gas and oil, Jordan faced mounting economic pressure in the early 2000s. By 2005, the cost of shielding domestic consumers, power producers, and industry from international fuel price hikes had reached an unsustainable 5.8 % of GDP—about USD 711 million annually (Bridle et al., 2014). In response, Jordan implemented a seven-year plan to completely remove fossil fuel subsidies. This reform was its third attempt in 30 years, following two previous efforts that failed due to public backlash (Fattouh & El-Katiri, 2012). Several factors contributed to the success of this reform. The government announced the changes well in advance and phased them in gradually. It also launched public information campaigns to explain the necessity of the changes and implemented social compensation measures. These included increasing public wages and pensions—particularly for low-income groups—and expanding food subsidies and social safety nets, benefiting nearly 60 % of the population. Importantly, the subsidy reform was part of a broader overhaul of the energy sector, which included restructuring the electricity system and charging consumers the true cost of power. However, Jordan did not introduce greener energy alternatives alongside the reforms. While fuel consumption and CO_2 emissions temporarily decreased, there was no simultaneous development of domestic green industries. Since 2009, emissions have been on the rise again (Merrill et al., 2015). This case illustrates that while removing fossil fuel subsidies can reduce energy consumption and create space for renewable energy investment, it is not sufficient on its own to drive a green transition. In Jordan’s case, the absence of industrial policies to actively promote renewable energy manufacturing or address structural barriers—such as financing, regulation, and institutional capacity—meant that green industries failed to emerge. *“The reform levelled the playing field but did not build the runway”*, to use the words of the author.

2.6 International spillovers, and the global dimension of GIP

One increasingly relevant aspect of green industrial policy (GIP) is its global dimension, particularly how cross-border spillovers and international trade rules interact with national policy efforts. Goldberg et al. (2024) highlight that in **nascent** sectors with strong learning effects and cross-border externalities—like semiconductors in Korea, Taiwan and China—state subsidies may generate positive international spillovers. These effects come from technology diffusion, learning-by-doing effects, foreign direct investment (FDI), research collaborations, and global supply chain integration, which benefit even countries not directly subsidising the industry (Goldberg et al., 2024, NBER).

When applied to green hydrogen, this insight has major implications. Clean hydrogen projects face significant uncertainties in many interrelated areas (systemic risk): location and availability of supply and demand, infrastructure readiness, pricing, regulations, etc. Given these barriers, countries are developing national hydrogen strategies not just for domestic reasons but to position themselves in a future global hydrogen market. For instance, the EU, Japan, and Korea’s hydrogen import targets (OECD, 2025) offer a credible signal to investors and exporters that demand will materialize—reducing risk and improving investment visibility. These policies do not only benefit the countries implementing them, but also other countries consuming green hydrogen. If countries succeed at scaling up the hydrogen industry, the technology cost declines and infrastructure knowledge generated can spill over globally, just as learning in one country’s chip factories can lower prices for international clients. These dynamics might suggest that green hydrogen, like semiconductors, may have “public good” characteristics at a global level.

However, Goldberg et al. also warn that spillovers are not automatic. Firms and countries can choose to share or restrict technology. In the case of green hydrogen, this could take the form of export controls, local content requirements, or limited technology licensing, all of which may restrict diffusion. Thus, a “green subsidy race”, if not carefully coordinated, risks becoming wasteful or protectionist—especially if it results in duplicated investment or fragmented standards. This raises a crucial question for global climate policy: how to ensure that national green industrial policies are designed to promote international cooperation, not conflict? The example of semiconductors suggests that multilateral frameworks, transparency, and international R&D collaboration are key. For hydrogen, efforts such as the International Partnership for Hydrogen and Fuel Cells or the EU’s “Clean Hydrogen Alliance” or other [bilateral agreements](#) may serve as a foundation for this kind of cooperative architecture.

Box 2. Insights from the sector of Electric Vehicles: *Implementation matters*

Goldberg (2025) highlights the complex effects of industrial policy using the example of electric vehicles (EVs), where nearly all EV-producing countries offer consumer subsidies. In China’s case, these subsidies were combined with a domestic content requirement, which had important implications for the global supply chain. The EVs supply chain is a particularly interesting example because a major cost driver of EVs is the electric battery, and in the early years of EV manufacturing (around 2015–2016), battery suppliers were mainly based in Japan and South Korea, producing at a cost lower than other suppliers at the time. At that time, the EV battery industry was still fragmented, with Japanese and Korean battery producers often supplying multiple EVs producers worldwide. Importantly, the learning-by-doing externalities in this industry occurred mostly in the battery production segment, rather than in car assembly itself (assembly was mature; cumulative production already huge). As Goldberg notes, China’s consumer subsidies increased domestic demand for EVs, which in turn boosted demand for batteries. Higher production volumes of batteries led to cost reductions in Japanese and Korean battery manufacturers due to learning-by-doing effects and economies of scale. This in turn lowered costs for EV carmakers in other countries—such as General Motors—which sourced batteries from those suppliers. However, between Jan 2016 and June 2019, China introduced a local content requirement (the “white list”), making subsidies available only for EVs equipped with batteries produced by Chinese firms. This directed demand towards domestic suppliers, allowing them—not foreign firms—to benefit from scale and learning advantages while Japanese and Korean producers lost market share. While such a policy supported China’s strategic goal of building a competitive battery industry (which worked markedly well), it reduced the positive international spillovers and learning-by-doing benefits that might have otherwise occurred in a more open supply chain.

Goldberg’s analysis provides valuable insight for the emerging green hydrogen sector, which may follow a similar supply chain structure. Green hydrogen production cost is driven heavily by the expensive electrolyzers, which are currently manufactured by a [limited number of countries](#) (China, Europe, etc.). If consumer subsidies for green hydrogen are introduced without domestic content requirements, they could increase global demand

for electrolyzers, triggering learning-by-doing effects and pushing costs down for all users internationally. However, if countries condition these subsidies on the use of locally produced electrolyzers the positive spillovers across borders will be diminished (each country may replicate China’s battery strategy advancing national champions but slowing collective progress). This highlights a critical point regarding industrial policy: the way industrial policy is implemented matters. As Goldberg (2025) argues, consumer subsidies—when designed without restrictive local content rules—can act as global complements, encouraging a “race to the top” where all producers benefit from cost reductions along the value chain. In contrast, if subsidies are tied to domestic inputs, they may entrench fragmentation and reduce collective progress. This dynamic is especially relevant as green hydrogen policy frameworks are still being formed and global supply chains are taking shape.

Sources: Markus’ Academy 2025. Pinelopi Goldberg on Industrial Policy in the Global Economy.

2.7 Should developing countries engage in green industrial policy?

The challenges faced by developing countries in implementing industrial policy are fundamentally different from those of advanced economies. In most cases, the sectors targeted in developing countries are traditional, not innovation-driven. As such, they do not typically generate significant learning externalities, which limits the standard justification for industrial policy based on knowledge spillovers. Nonetheless, there are still strong arguments in favour of targeted industrial policy in the developing world. Reed (2024) argues that export-led industrial policy is more appropriate in these contexts. Given their limited fiscal space, developing countries cannot afford broad, economy-wide interventions and must instead focus their efforts on a few promising sectors. This raises the question of how to select these sectors. While governments may lack the ability to “pick winners,” Reed suggests that they can at least avoid picking losers by applying objective criteria such as: International market growth potential; Level of global competition; Existing comparative advantage (*see Box 3*); Technological relatedness to existing domestic capabilities. Importantly, Reed emphasizes that developing countries must adhere to the principles of multilateralism. Because unlike the U.S., China, or the EU—which can afford to bend or even break global trade rules—developing countries must remain pragmatic and operate within the established international framework.

A good example is Costa Rica, which successfully transitioned from being a traditional exporter of bananas and coffee to a key player in medical devices and business services, thanks to an industrial policy that attracted major investors like Intel. However, Reed and others caution that copy-paste policies should be avoided, as countries blindly replicating each other’s strategies risk engaging in a “race to the bottom.” This emphasises again the importance of international coordination and rule-making, which becomes especially critical for low- and lower-middle-income countries. As Goldberg (2025) notes, the risk of a race to the bottom is more acute in traditional sectors where competition is high and technological spillovers are limited. By contrast, this issue becomes less pronounced in innovative sectors, such as green hydrogen, where high upfront costs and learning externalities create positive-sum dynamics. In such cases, subsidy competition can lead to mutual benefits rather than zero-sum rivalry—especially when international spillovers are allowed to occur.

Box 3. The commercial rationale for emerging economies in exploring and exporting green hydrogen: The case of Mongolia

The policy literature on promoting investment in green hydrogen—especially from the private sector—emphasises the high-risk nature of this emerging industry. Developing clean hydrogen requires large-scale capital for infrastructure such as electrolyzers, renewable energy systems, and facilities for transport and storage. At the same time, the sector is still developing and is affected by multiple overlapping uncertainties, including supply and demand locations, infrastructure development, pricing, and regulatory frameworks. Because of these risks, governments in countries with strong ambitions for hydrogen use are actively looking for ways to increase predictability and lower investment risks. In recent years, many have introduced national hydrogen strategies aimed at supporting this effort. For instance, the EU, Japan, and South Korea have announced import targets that collectively reach almost 16 million tonnes by 2030 (OECD, 2025).

Globally, countries are designing and implementing policy programmes to support the decarbonisation of their economies, in line with the Paris Agreement—to which Mongolia is a signatory—which aims to limit global temperature increases to 1.5°C above pre-industrial levels by the end of the century. The main challenge for all countries pursuing this goal lies in the complexity of coordinating the low-carbon transition, which increasingly requires governments to take an active, interventionist role in driving systemic change—often through a renewed interest in industrial policy (Criscuolo et al., 2022). Green hydrogen is one of the few technologies that can support the decarbonisation of both Mongolia's domestic industrial sectors and offer the country an opportunity to pursue an export-led industrial policy. In the mining sector, which is expected to be the initial source of domestic hydrogen demand, hydrogen can power heavy-duty vehicles where electrification is not viable (Nilsson et al., 2021).

While technological innovation in green hydrogen production is currently concentrated in a few advanced countries—mainly Japan, the United States, Germany, and Korea—many emerging economies, including Mongolia, are well-positioned, due to their abundant renewable energy resources and available land, for large-scale green hydrogen production (Cammeraat et al., 2022). According to the IEA (2023), global demand for low-emissions hydrogen is projected to grow from 1 Mt in 2022 to 420 Mt by 2050, with 327 Mt expected to be produced using electrolysis powered by renewable energy. The levelised cost of green hydrogen (LCOH) depends heavily on two factors: the cost of electricity and electrolyzers. As the cost of renewable electricity is expected to come down in the coming years, it will likely become the main determinant of LCOH over the medium and long term. In this context, Mongolia—endowed with vast land, high solar irradiance, and high wind speeds—could have a significant comparative advantage in green hydrogen production, provided it adopts the right green industrial policies to attract and scale investment in its renewable energy sector. If Mongolia succeeds in producing low-cost green hydrogen, it could serve rapidly growing markets in neighbouring regions, particularly northern China, where steel manufacturers are increasingly investing in hydrogen to maintain competitiveness in a low-carbon economy (Hseuh, 2023; Zhou et al., 2022).

3 Methodology

3.1 About the data

To assess the effectiveness of various industrial policies supporting green hydrogen deployment across countries, I created a dataset that groups and categorizes relevant policies implemented or announced since September 2023. The primary objective was to identify similarities among policies and classify them based on their key features. To do this, I applied an unsupervised machine learning technique—Agglomerative Clustering—which is well-suited to small and medium-sized datasets (see section 3.3 below). The resulting dataset comprises 107 individual policy entries across 33 countries/jurisdictions and includes eight variables: *Country* / *Status* / *Sector* / *Programme* / *Instrument_type* / *Policy_side* / *Amount_USD_mn* / *Duration_yrs* / *Description*.

Data collection focused exclusively on official sources, including international organization reports and national government documentation. The IEA’s Global Hydrogen Review 2024 was the most comprehensive and frequently cited source, especially for policy announcements and cost estimates. According to the IEA, there are now more than 2,070 planned green hydrogen projects worldwide, underscoring the sector’s rapid expansion and the increasing need for supportive policy frameworks (IEA, 2024).

To ensure consistency and reliability, the dataset includes only policies that are either “In Force” or “Announced” with a clearly defined framework and, where possible, a cost estimate. The *'Status'* variable is a binary variable indicating these two conditions. The *'Sector'* variable captures the specific area targeted by the policy, such as aviation, hydrogen production, electrolyzers, or more generally, stages in the innovation cycle like R&D or technology demonstration. The *'Programme'* column identifies the broader programme under which the policy falls (e.g., Hy2Use IPCEI in the EU, or the Inflation Reduction Act in the US). The *'Instrument_type'* variable classifies the policy tool being used (e.g., grant, tax credit, Contracts for Difference (CfDs)). The *'Policy_side'* variable is a derived binary indicator categorizing whether the policy is a Push or Pull measure. Push policies typically aim to de-risk investment and production through upfront support (e.g., grants), while Pull policies aim to ensure future revenue stability (e.g., CfDs). Classification was determined through qualitative analysis (i.e., close reading of policy descriptions from official reports). In two cases—policies from Australia and Denmark—where both push and pull elements were present, classification was based on the policy’s dominant features to preserve the binary state of the variable. Future updates shall introduce a third category within the variable to reflect hybrid policy designs. The *'Amount_USD_mn'* variable captures the estimated financial commitment in millions of USD. Currency conversions, when needed, were done using the 2024 IMF exchange rates for consistency with the IEA 2024 report. The *'Duration_yrs'* variable reflects the time frame over which each policy is intended to be implemented, where this information was available. Lastly, the *'Description'* variable provides qualitative detail about each policy’s structure, design, and objectives. [Click here to download the dataset.](#)²

²The linked dataset contains two extra columns i.e., *cluster7* and *cluster3*. These two columns resulted from performing the clustering analysis and were therefore added later. (see section 3.3)

3.2 Insights from the dataset

Figure 1 shows where different hydrogen-policy instruments are being used around the world. Each coloured dot represents one instrument (a grant, a tax credit, a CfD, etc.). The size of the dot grows when a country uses that instrument many times.

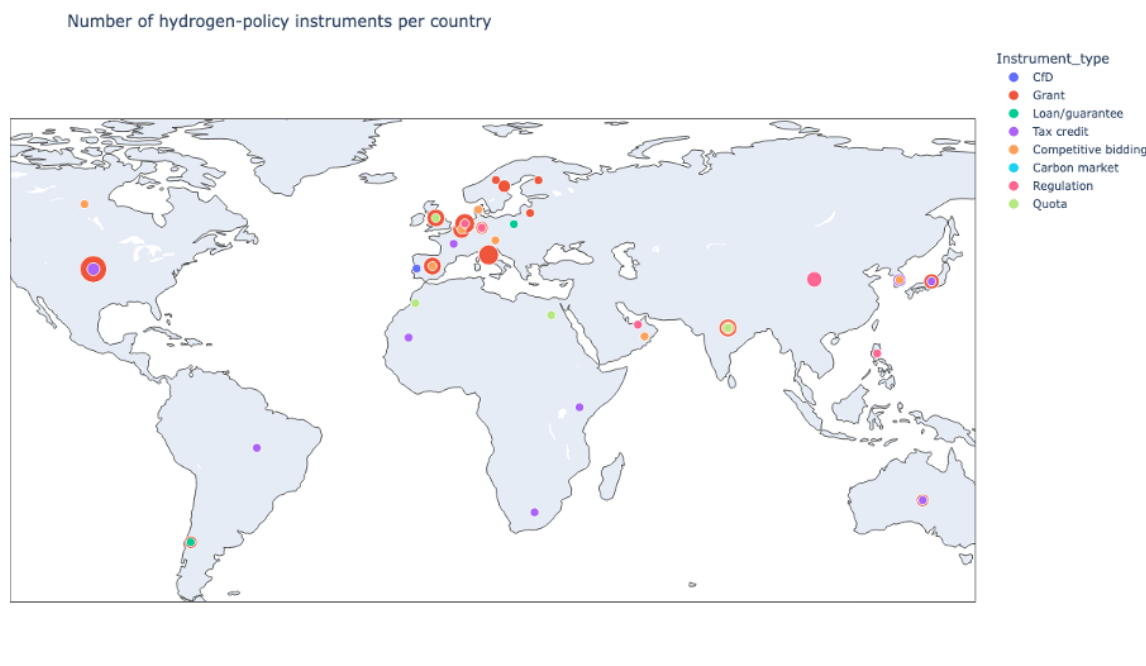


Figure 1: Global map of hydrogen-policy instruments by type and frequency, since September 2023.

From the map we can see three clear stories: (1) Multi-instrument policy hubs; United States and Western/Northern Europe (Germany, France, the Netherlands, the UK, Denmark . . .) show the repeated usage of many instruments simultaneously (size of the dot)³. In the U.S. the main instruments are grants and tax credits created by the Inflation Reduction Act. In Europe the colours mix: grants, Contracts-for-Difference (CfD) pilots, and competitive bidding auctions such as the Auction-as-a-service scheme⁴ in several European countries which simplifies the auction process by using a standardised design for all member states (The European Commission, 2024). These regions are testing several tools at once—grants to kick-start projects and market-based schemes to control long-term costs. (2) Regulation-centred countries; Japan, South Korea and China show mainly regulations or quotas and a few competitive bidding schemes. Rather than giving out large subsidies, these governments prefer to set rules or targets—for example, the Chinese National Development

³The support announced by the European Commission was grouped under the name 'Belgium' for the sake of creating the map; the *plotly.express* python function iterates over an inherent list of country names which need to match the one in the dataset in order to work.

⁴This [model](#) simplifies the auction process by using a standardised design for all member states and the same tender process to enable the funding of additional projects.

and Reform Commission issued actions plans for refining and ammonia industries which explicitly mention use of green hydrogen, any by the end of 2025, any capacity below a defined threshold will have to be improved or shut down (IEA, 2024). (3) Emerging and exploratory markets; In Latin America, Africa, South-East Asia and the Middle East we see only a handful of dots, almost all tax credits. [Click here to access the interactive map](#); click twice on the name of an instrument to isolate the usage of an instrument per country.

On the typology of instruments: Grants dominate the *Instrument_type* column, accounting for 52 of the 77 policies that report an estimated budget. The remaining measures reporting budgets are far less common: competitive-bidding schemes appear nine times, contracts-for-difference seven times, tax credits five times, and loan/guarantee facilities four times. These are the only five instrument classes for which reliable cost estimates were available. Therefore, they will constitute the full analytical sample used in the *Amount_USD_mn* variable, which will be used in the clustering analysis later. Table 3 below summarises and compares these five instrument types, highlighting their relative frequency and distinguishing design features within the broader landscape of renewable-energy support.

Instrument	Main Countries of Deployment	Common Policy Contexts	Pros	Cons
Grant	Widely used across North America, Europe, and Asia. The dataset counts United States (9 grant GH programs), Netherlands & Italy , four in India .	Early-stage R&D and demonstration, pilot plants, and infrastructure build-out before commercial maturity; a classic <i>push</i> mechanism for innovation. (1)	Direct targeting of strategic technologies (1); removes up-front capital barrier; spurs private R&D by sharing risk (2); generally transparent and auditable (2).	Budget-intensive for governments; slow, bureaucratic award processes; risk of misallocating funds (“picking winners”) if selection is poor (2).
Tax Credit	Dominant in the United States (IRA’s ITC/PTC, 48Q/45V hydrogen credits) and seen in South Korea, Canada, Brazil .	Deployment-stage support that lowers effective project cost by reducing tax liability (2); works best where firms have sufficient taxable income.	“As-of-right” incentive speeds uptake; mobilises large volumes of private capital; perceived as market-friendly and politically palatable (2).	Limited targeting and transparency (2); potential free-riders; reliance on tax-equity financing raises transaction costs and can exclude smaller developers.
Competitive bidding	Common across Europe, Asia, and Latin America; dataset includes India, Brazil, Germany, Spain, UK . By 2024, over 100 jurisdictions run renewable auctions (3).	Procures large-scale capacity once technologies mature; competitive tender reveals market price and allocates subsidies efficiently (4).	Transparent price discovery drives costs down; governments control volume and spending; winning projects secure long-term PPAs/CfDs, improving profitability (3).	Aggressive competition can lead to under-bidding which can delay delivery; high bid-preparation costs if not enough bidders; requires robust oversight to prevent collusion.
Loan/ Guarantee	Dataset shows programmes in Australia, Canada, Chile, Japan, Poland ; major schemes also in the USA (DOE LPO), Germany (KfW), UK (UKIB), China .	De-risks debt for first-of-kind or capital-intensive projects; government loans or guarantees lower interest rates and unlock commercial finance (5).	Leverages private capital while limiting near-term public outlay; significantly reduces cost of capital; successful repayments repay loans (5).	Taxpayers assume default risk; moral hazard if firms take excessive risk; complex due-diligence and monitoring requirements; potential politicisation (6).
CfD	First scaled in the UK (offshore wind, 2014) and now adopted or piloted in Germany, France, Netherlands, Italy, Japan, Australia (esp. for hydrogen).	Used mainly in electricity markets seeking to hedge merchant-price risk; strike price agreed via auction, paying/recouping the difference to market price.	Provides long-term revenue certainty, lowering financing costs; symmetric claw-back protects consumers when market prices exceed agreed strike price (7).	Can mute market-price signals (output generation even at negative prices); long-term fiscal liability; requires sophisticated market and fiscal capacity to implement (7).

Table 3: Comparative Analysis of Renewable-Energy Policy Instruments (for Green Hydrogen and beyond). *Numbers in parentheses refer to sources.*

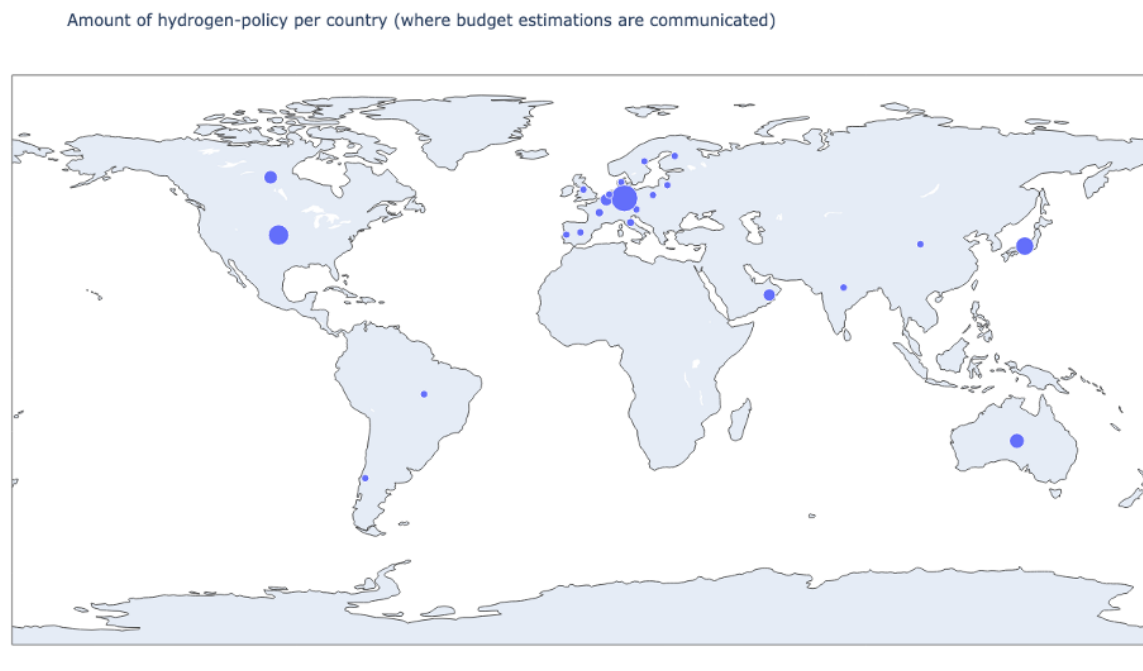


Figure 2: The cost of green-hydrogen support policies entering into force since September 2023

In Europe, Germany stands out with the most significant policy package, largely driven by the launch of its H_2 Global off-take scheme and updated funding guidelines for electrolyser deployment. France, the Netherlands, Spain, and the United Kingdom show medium-sized policies, reflecting updates to national hydrogen strategies that include instruments like feed-in premia, CfD-style auctions, and CAPEX grants. In North America, the United States holds the second-largest share of global green hydrogen policy support, led primarily by the Clean Hydrogen Production Tax Credit introduced under the Inflation Reduction Act (Section 45V)⁵. Canada also shows an important—though smaller—policy budget, reflecting the Clean Hydrogen Investment Tax Credit alongside several provincial procurement measures that took effect in early 2024. In emerging economies such as Brazil, Chile, and Oman, green hydrogen policies are generally centred around tax credits and demonstration plant funding. In Chile’s case, additional instruments include loans and partial guarantees for project developers, reflecting early-stage public support in newer hydrogen markets (IEA, 2024). [Click here to access the interactive map.](#)

⁵There are several budget estimations for this tax credit (some go up to USD 100 billions). The estimation used in the created dataset is the revised estimation from the Congressional Budget Office (CBO) and the Joint Committee on Taxation (JCT) at around USD 20 billions.

Box 4. The Neom Project and the Arabian Gulf region

There are also several flagship policies aimed at accelerating the roll-out of green-hydrogen production in the Arabian Gulf region. For instance, Oman’s competitive bidding schemes to award land to two projects worth USD 11 billion in investment, which are to exceed its target of 1 Mtpa H_2 2030 target (reaching 1.4 Mtpa H_2). Or the UAE’s strategy which forecasts a production of 7.5 Mtpa H_2 by 2040 and 15 Mtpa H_2 by 2050 (IEA 2023; IEA 2024).

Another chief example is Saudi Arabia’s NEOM Green Hydrogen and Ammonia Project. Since 2023, the Saudi government has channelled USD 2.75 billions of concessional finance (i.e., below-market-rate financing) through the National Development Fund and the Saudi Industrial Development Fund to support construction. The project is now backed by a wider USD 8.4 billion financing package arranged with 23 international lenders, making it one of the largest green-loan deals to date:

- **Scale and technology:** Located at Oxagon in the NEOM economic zone, the plant will integrate about 4 GW of dedicated solar and wind power to run 2.2 GW of alkaline electrolyzers—currently the world’s largest electrolyser installation under construction. Expected output is roughly 600 t H_2 per day, converted on-site into 1.2 million t per year of green ammonia for export.
- **Timing and impact:** Ground works are complete and major contracts are awarded; commissioning is scheduled for late-2026. Once operational, the plant is expected to avoid about 5 million t CO_2 equivalent each year by displacing grey-ammonia and diesel use in heavy transport and industry.
- The project is a centre-piece of Saudi Vision 2030 and the Saudi Green Initiative, positioning the country as a first-mover exporter in the emerging green-hydrogen market while providing a replicable template for gigawatt-scale projects elsewhere.

Sources: Neom case study OECD and neom.com

3.3 Empirical strategy

To identify patterns in the design of green hydrogen support policies, I use Hierarchical Agglomerative Clustering (HAC)—an unsupervised machine learning technique that iteratively groups observations based on their similarity (*see Box 5*). Unlike classical econometric models, which are parametric and therefore impose a specific functional form on the relationship between dependent and explanatory variables (e.g., a linear relationship in OLS regression models, or a linear relationship through a link function in logistic regression), most machine learning algorithms—including HAC—are non-parametric: they make few a priori assumptions about the underlying data-generating process (Azzopardi et al., 2019). This flexibility often leads to higher predictive accuracy but also increases the risk of overfitting and typically requires a large data structure to generalize well (Bazarbash, 2019; Jordan & Mitchell, 2015).

Machine learning methods are conventionally divided into *supervised* and *unsupervised* cate-

gories. Supervised algorithms learn a mapping from features to a labelled outcome variable—for example, an airline company trying to predict new passengers’ satisfaction using historical survey scores from previous passengers, or a bank attempting to forecast customer churn based on past account data from other customers. In contrast, unsupervised algorithms operate without labelled outcomes; their aim is to discover hidden structure in the data itself. A common example of unsupervised machine learning would be a marketing programme that sends different promotional emails to different customers, depending on their engagement patterns with the brand. When the clustering model receives a new customer, it assigns them to the most appropriate cluster (i.e., marketing group), so they receive the most relevant promotion. Therefore, both OLS and logistic regressions—when seen through the lens of machine learning—are considered supervised learning models. And because no reliable outcome variable of *green hydrogen GIP effectiveness* is available in the dataset (e.g., a binary variable indicating whether a policy was effective based on a green hydrogen production threshold, or a continuous variable reflecting CO_2 -equivalent emissions avoided per tonne of green hydrogen produced), a supervised learning approach is not appropriate. In this context, an unsupervised method such as clustering is more suitable. Clustering partitions observations into internally coherent, externally distinct groups, offering an empirical basis for generating typologies of policy design.

Among unsupervised clustering methods, HAC offers several advantages. First, it is well suited for small-to-medium-sized datasets, where the issues of high dimensionality are less severe, and where the data may not meet the spherical cluster assumption required by methods like k-means clustering (James et al., 2013). Second, HAC produces a dendrogram—a tree-like visual representation of nested groupings—that aids in qualitative interpretation. Third, HAC does not require the number of clusters k to be specified in advance; the number can be chosen *ex post* by visually inspecting the dendrogram and selecting a cut-off point based on the height of linkages. For this study, the model uses a linkage method with Euclidean distance, which minimizes the increase in within-cluster variance at each merge step and tends to produce compact and well-separated clusters. Before applying the clustering algorithm, all seven numeric features were standardised to have zero mean and one unit standard deviation, ensuring that no single variable dominated the distance calculations. While budget data is essential for understanding the scale of policy instruments, around 27 % of policy entries lacked budget estimates (insight from the dataset). To maintain comparability, the dataset was filtered to include only observations with non-missing budget values, reducing the sample to 77 policies. This filtering step also reduced the number of distinct instrument types from eight to five, corresponding to those discussed in Table 3.

Applying HAC to the cleaned dataset resulted in seven statistically distinct clusters. The associated dendrogram (*see Figure 3*) shows a pronounced increase in linkage distance at the seventh merge, suggesting this as the natural cut-off. The tight clusters at the lower levels of the dendrogram further reflect strong internal coherence among the grouped policies.

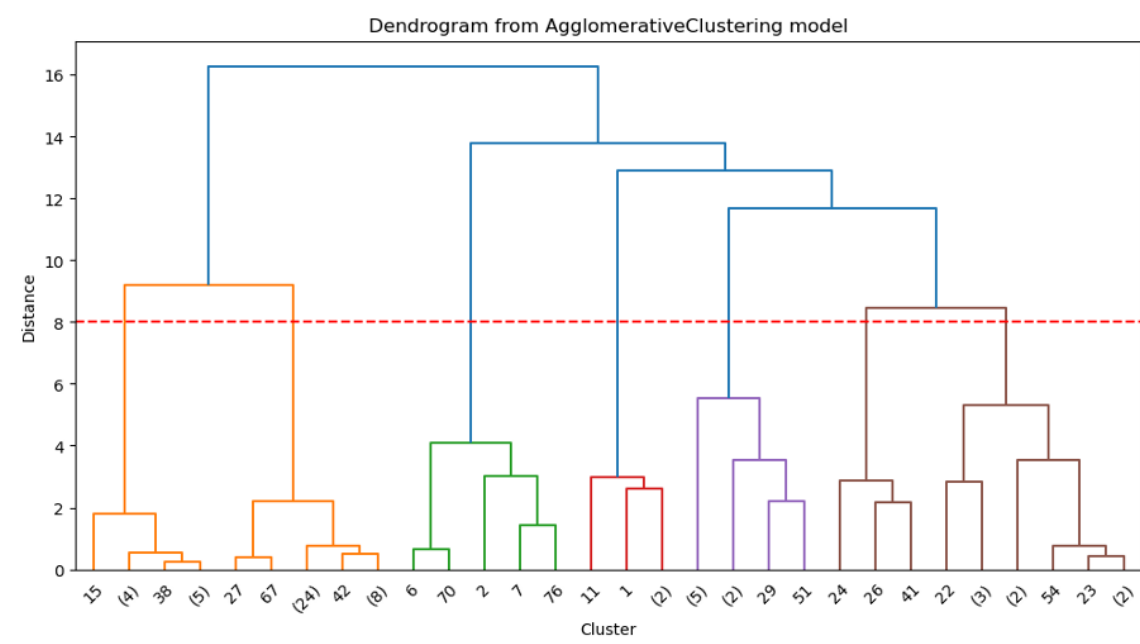


Figure 3: Dendrogram showing the resulting clusters, the cut-off is done at the intersection with the horizontal line at 8, (i.e., 7 natural clusters)

To better visualize the structure of the seven clusters, I performed a Principal Component Analysis (PCA) and projected the data onto two dimensions. While the actual clustering was conducted in the full seven-dimensional feature space, this two-dimensional PCA projection—shown in Figure 4—still provides valuable insight into the distinctiveness of the clusters. The plot reveals clear patterns: for instance, Clusters 4 and 5 form well-defined and compact groupings, suggesting strong internal coherence. Meanwhile, Cluster 0 appears to share partial similarities with multiple other groups, indicating potential overlap in policy features. Although some dimensionality is lost in the projection, the visualization offers an intuitive snapshot of how policies are distributed across the identified design space.

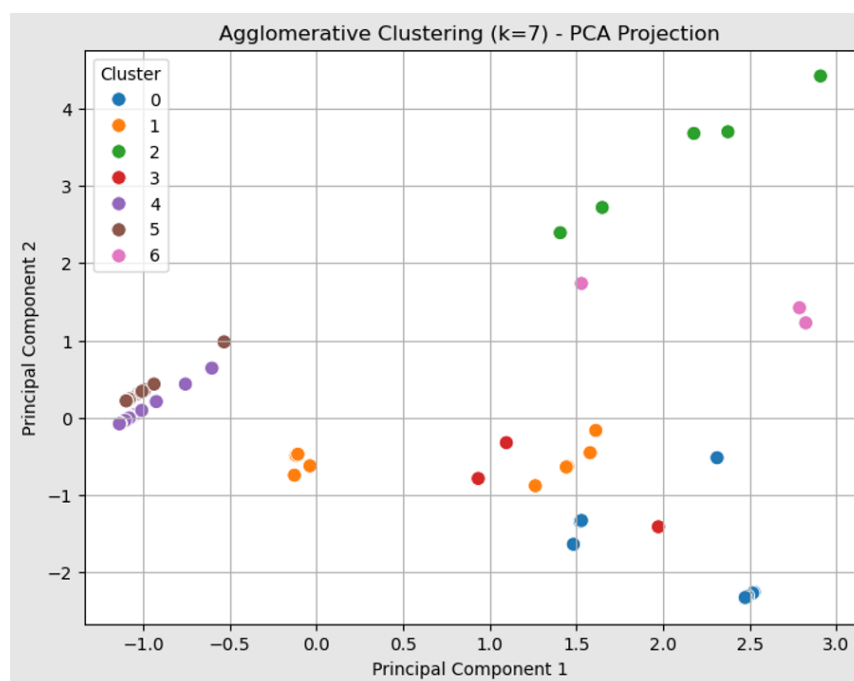
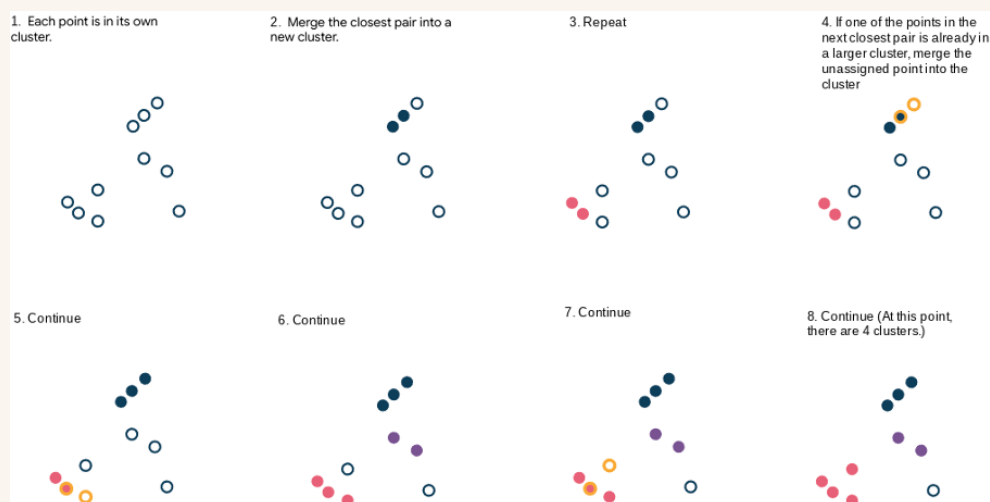
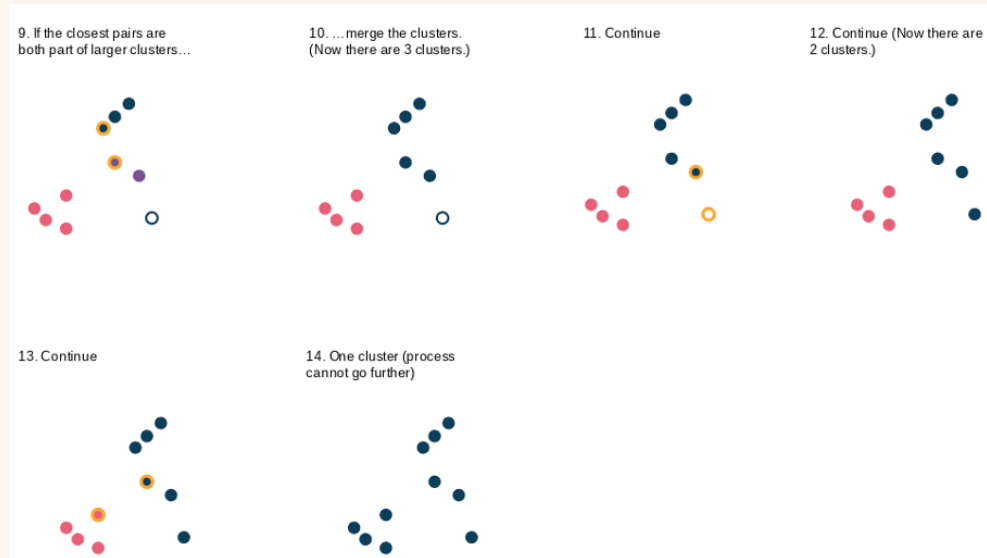


Figure 4: Principal Component Analysis (PCA) to better visualise the clusters

Box 5. Agglomerative clustering

One way to cluster the data is by using a technique called agglomerative clustering. Agglomerative clustering works by first assigning every point to its own cluster, then progressively combining clusters based on inter-cluster distance. Here's how it works:





Agglomerative clustering requires that you specify a desired number of clusters or a distance threshold, which is the linkage distance above which clusters will not be merged.

Source: Google Advanced Data Analytics Professional Certificate, 2023

The first intuition guiding the modeller is to expect five clusters, each corresponding to one of the five distinct policy instrument types included in the dataset. Figure 5 shows the distribution of policy instruments across the resulting clusters. Interestingly, both clusters 6 and 1 contain a mix of grants and Contracts for Difference (CfDs)—this might indicate a degree of similarity between these two instruments which will be revealed in the subsequent differentiation by instrument and policy side. Overall the agglomerative clustering analysis reveals several valuable insights. Most notably, the model identifies that policy data points are broadly distinguishable by instrument type, and that grants are by far the dominant policy instrument in the current green hydrogen policy landscape.

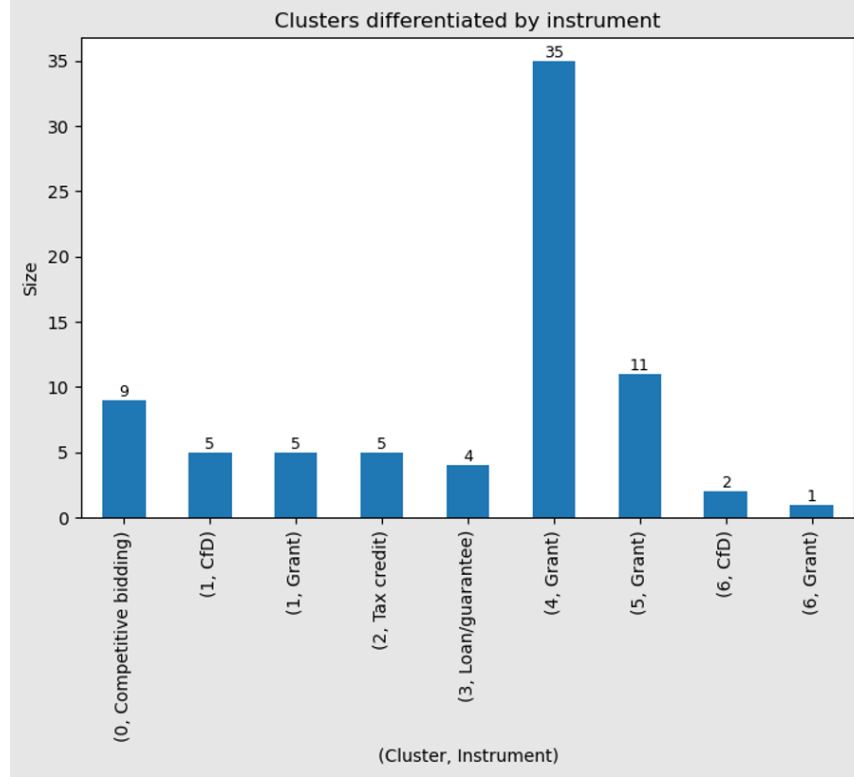


Figure 5: Resulting clusters differentiated by type of instrument

To better understand why the model produced seven clusters despite only five instrument types, I explored the clustering results using an additional policy feature: the policy target, or *Policy_side* variable. This binary variable distinguishes between push-type instruments (which support early-stage project development and de-risk investment) and pull-type instruments (which stabilize or enhance revenue post-implementation). This multidimensional analysis (*see Figure 6*) yielded an important insight: clusters can be differentiated not only by instrument type but also by policy direction. For instance, grants are highly differentiable depending on whether they function as push or pull instruments. In cases where grants are grouped in the same cluster as CfDs (e.g., Cluster 1), they appear to serve a pull function, aligning well with the revenue support typically offered by CfDs. Meanwhile, tax credits are entirely concentrated in a single cluster, reflecting their relatively homogeneous push usage across countries. This pattern is consistent with the current global practice—to date, there are no known tax credit schemes targeting consumers in green hydrogen markets. Furthermore, the scarcity of demand-side policies is a salient outcome of the clustering exercise. With the exception of competitive bidding mechanisms (clustered in Cluster 0), the supply–demand balance in green hydrogen policy appears heavily skewed toward supply-side support—particularly through grants. One possible interpretation is that the policy ecosystem is still in the early stages of market formation, where governments prioritize building supply capacity before engaging in demand-side interventions. Alternatively, governments may be responding with ‘tit-for-tat’ mechanisms, replicating each other’s policies.

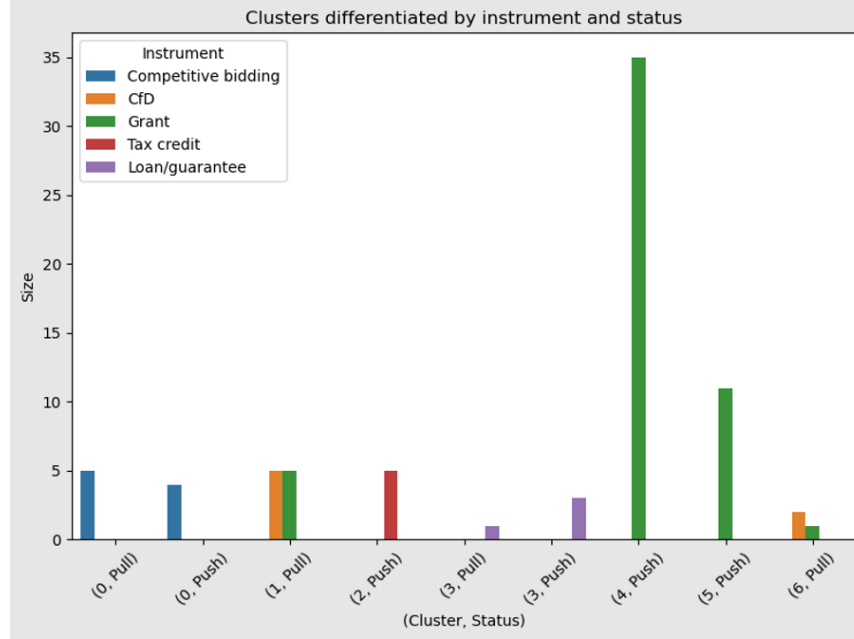


Figure 6: Resulting clusters differentiated by type of instrument *and* policy target

These findings support the robustness of the clustering model at hand. What is more, the results were tested on different modelling varieties (i.e., different tuning of hyper parameters) and with a k-means clustering model⁶, the result were nearly identical in all trials which provides further evidence on the visible underlying structure of the data.

How is this helpful? The use of agglomerative clustering in the context of green hydrogen policy analysis is beneficial for several reasons. First, it confirms the robustness of the dataset, which is encouraging and supports continuing to expand the dataset as new policies are announced. Second, being able to assign new data points to existing clusters is valuable when using a model trained on a larger dataset that needs to process new information as it becomes available. For example, as green hydrogen markets develop further and governments introduce more support measures, the model can assign each new policy to one of the existing clusters based on its design features. This can provide early insights into the policy’s likely behaviour or effectiveness. Third, both the cluster assignments and the distance to cluster centroids (i.e., the mean of the cluster) can be used as new input features in regression or supervised machine learning models. For instance, we can take the training data of a supervised learning problem, apply clustering, and then use either the cluster labels or the distance matrices as predictor variables. This process can help improve the predictive power of models like OLS regression or logistic regression or others, especially when trying to predict outcomes such as policy success or emissions impact.

To further understand the structure of green hydrogen policies, I set the model to form three

⁶A straightforward clustering technique which minimises the residuals between the centroid of a cluster and the surrounding data points (*see Box 5*).

clusters to explore what insights could be drawn, especially in relation to policy type and country characteristics. The resulting clusters (0, 1, and 2) still reflect groupings based on instrument type and policy direction. As shown in Figure 7, supply-side grants and tax credits—which also support supply—fall into separate clusters. Meanwhile, Cluster 0 includes demand-side policies as well as some competitive bidding and loan/guarantee push instruments. These may have been grouped together due to shared features like policy size or other design characteristics. This simplified clustering still offers useful distinctions in how different instruments are used across countries.

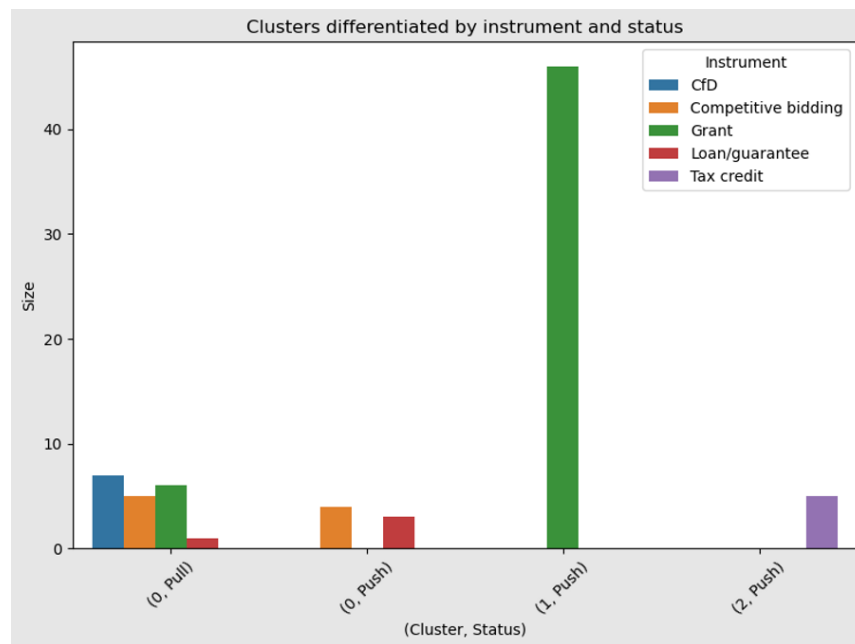


Figure 7: 3 clusters (0, 1, 2) differentiated by type of instrument and policy target

To check whether clustering the data into three groups reveals any patterns by country type, I created three heatmaps—one for each cluster—showing the countries represented and their preferences for specific types of policies. Each heatmap is indexed by country, making it easier to identify regional trends and see which types of support are more common in different areas. The analysis produced the following three maps:

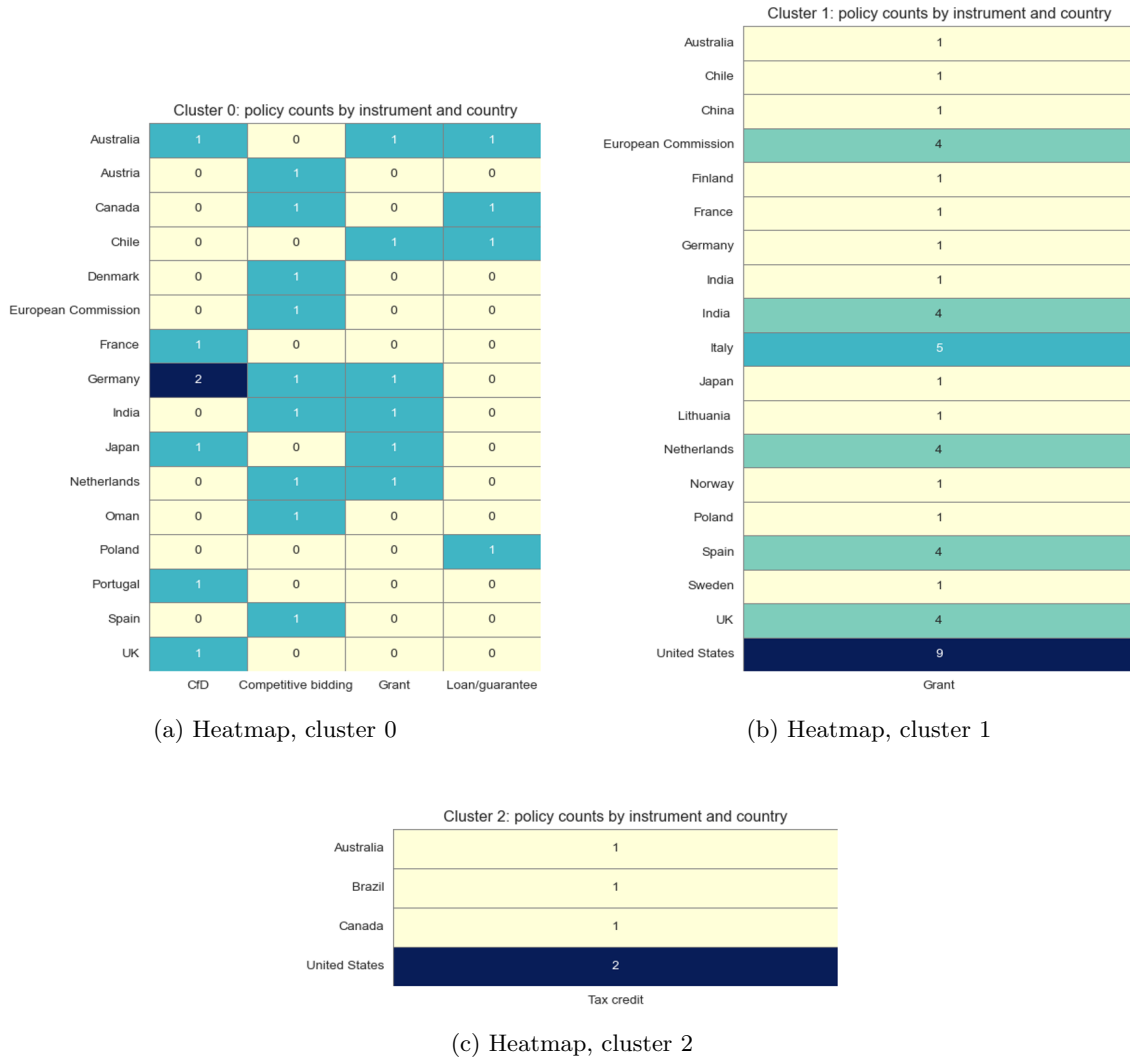


Figure 8: Cluster-differentiated heatmaps illustrating the usage/preference of instrument type per country

The three heatmaps above contrast the composition of the three model clusters along policy instrument and country of origin—and shows that the groups are defined far more by *how* governments intervene than by *where* those interventions are drafted. Cluster 0 is a genuine mixed-instrument toolbox. Contracts-for-difference, competitive-bidding schemes, grants and a few loan-or-guarantee programmes each appear, but never in large numbers; most countries contribute only a single measure of any given type. Sixteen jurisdictions from Europe, Asia, Oceania and the Middle East are represented, with none supplying more than two policies. This wide geographic spread and low within-country density indicate that Cluster 0 captures the policy design rather than the regional distribution of policies. Cluster 1 is characterised exclusively by direct grant funding. Although

18 different countries still figure in the group, the contribution is uneven: the United States, Italy, the European Commission, the Netherlands and Spain together account for roughly two-thirds of the entries. Grants are therefore a globally used instrument, but the scale and frequency of their deployment are concentrated in a handful of mature hydrogen markets. Cluster 2 is the tax-credit niche. Only five policies fall into this cluster, drawn almost entirely from the Americas—two from the United States and one each from Canada and Brazil—plus a single Australian measure. This narrow instrument palette and regional focus suggest that the tax-credit approach is, at present, an Anglo-American preference rather than a universally adopted tool⁷. Thus, the heat-maps confirm that the unsupervised clustering algorithm primarily groups policies by their design features—mixed portfolio, grant-led, or tax-credit—while geography plays a secondary role. Countries such as Australia appear in every cluster, meaning that national portfolios often contain multiple design archetypes.

4 Results and limitations

One of the first striking findings from the data collection process on green hydrogen industrial policy is the imbalance between supply-side and demand-side interventions. According to the dataset built for this study, only about 22 % of announced policy instruments since September 2023 can be classified as demand-side (“pull”) policies (insight from the dataset). This suggests a heavy global bias toward supply-side (“push”) support, such as grants and tax credits, with relatively little focus on stimulating actual market uptake. A deeper look at the ‘Description’ column in the dataset shows many inconsistencies and gaps between the announced policy objectives and the actual instruments deployed. In several cases, the beneficiaries of the policies are unclear, and many entries lack information about expected production volumes, electrolyser capacity, or even a targeted sector. This raises concerns that many governments may be operating without a clear strategy, adopting green hydrogen policies opportunistically or reactively rather than with a coherent industrial vision. Another challenge arises from the difficulty of comparing policy types across countries. The design of green industrial policies varies widely—ranging from direct grants to below-market-rate financing (e.g., Saudi Arabia and China), land concessions, and technology-neutral decarbonisation subsidies. Many of these tools affect green hydrogen deployment indirectly, making it hard to attribute policy outcomes specifically to green hydrogen.

Moreover, this analysis is hindered by data limitations, particularly around three key issues: (1) Ex-ante vs. ex-post policy effects: Announcements are often made without clear implementation follow-through. For instance, large-scale policy announcements may fail to translate into actual uptake by firms unless production targets and financial mechanisms are specified (*see table 4 below*). Measuring the effectiveness of such policies requires tracking both the initial commitments (ex-ante) and their real-world outcomes (ex-post)—a challenging task in most countries. (2) Stock vs. flow: The dataset focuses on currently announced and active policies (as of post-September 2023), but this excludes past industrial policies that may have played a significant role in green hydrogen deployment. For example, in the case of China’s EV sector, although subsidies per unit have declined significantly since 2015, earlier generous support helped create globally competitive

⁷This claim is not very well-founded because different countries worldwide have announced tax credits (albeit without providing clear budget calculations). One might say Anglo-American preference until the moment we start having budget estimations about tax credits announced in other countries, and then they can be added to the analysis.

battery manufacturers (Goldberg, 2025). That historical context is difficult to integrate into policy comparisons. (3) Policy implementation complexity: In countries like China, the United States and Canada, green hydrogen policy is often announced at the national level but implemented at the provincial or state level, leading to coordination challenges and the risk of double-counting (IEA, 2024). This fragmentation complicates measurement and obscures policy intent.

Company	Objectives & Current Strategy (green H ₂)	Key Policy Instruments Leveraged	Public Support Received	Uptake / Flagship Projects (status 2025)
TotalEnergies	- Decarbonise hydrogen used in European refineries by 2030; Created green hydrogen business unit (2020) - Large green-H₂ projects (Masshylvia 40 MW; Normand’Hy 200 MW) - TEAL Mobility joint venture for heavy-duty transport	- France: National Hydrogen Strategy & “France 2030” (AMI calls) - EU: Green Deal & Hydrogen Strategy; IPCEI & Innovation Fund applications	€190m French state aid reserved for Normand’Hy (partnership with Air Liquide, awaiting EC clearance) Masshylvia submitted for IPCEI / Innovation Fund support	Masshylvia under construction → 5 t/day green H₂ from 2025 - TEAL Mobility joint venture (2024) rolling out 100 H₂ truck stations across FR/Benelux/DE
Air Liquide	“ADVANCE” plan makes H₂ a growth pillar (2022–25) - Full value-chain play: Siemens joint venture gigawatt electrolyser plant; Normand’Hy; - ENHANCE ammonia-to-H₂ project	- France: Hydrogen Strategy & France 2030; IPCEI Hy2Tech/Hy2Use participation - EU: €110m Innovation Fund grant (ENHANCE) + Green Deal framework	€190m in French grant earmarked for 200 MW Normand’Hy €110m EU Innovation Fund grant for ENHANCE (Antwerp)	Normand’Hy electrolyser online 2025 (supplies TotalEnergies refinery) - Berlin gigawatt electrolyser factory inaugurated 2023 → 3 GW/year - TEAL joint venture with TotalEnergies (100 H₂ stations)
ArcelorMittal	- Replace coal blast furnaces with H₂-DRI + EAF - Cut French CO₂ by 40% by 2030 (part of €1.7bn decarb plan)	France 2030 industrial strategy; EU Green Deal Industrial Plan / State-aid rules	€850m in French grant (EU-approved Jul 2023) to build 2.5 Mt/yr DRI & two EAFs in Dunkirk	- Construction 2023–27 - DRI & EAFs will shut two blast furnaces and cut 7.8 Mt CO₂/yr - FID steps continuing; timetable under review (Nov 2024 update)

Table 4: Overview of major industrial actors in France leveraging green hydrogen industrial policy
Sources: Air Liquide, TotalEnergies, Siemens Energy, European Commission, ArcelorMittal, IEA

From a methodological standpoint, the application of Hierarchical Agglomerative Clustering (HAC) revealed that industrial policies tend to be differentiated more by the types of instruments

used rather than by the country of origin. In the seven-cluster model, the grouping was driven largely by design features such as push/pull orientation, budget size, and instrument category. When the model was simplified to three clusters, the main distinction still rested on how countries intervene, not who intervenes. This suggests that despite national differences, policy makers often adopt similar tools, possibly due to governmental reactions or institutional learning.⁸ Visualizations, including heatmaps indexed by country and cluster, support this finding and sometimes highlight regional patterns in policy emphasis.

Europe and North America exhibit a concentration of policies linked to explicit climate or innovation goals, while emerging economies, when present, often emphasize infrastructure readiness or fossil-fuel substitution (insight from the dataset). Another important insight is the underdeveloped demand-side policy landscape. While many governments have introduced supply-side tools to reduce risk for producers, few countries have created strong pull mechanisms—such as carbon contracts for difference (CfDs), public procurement mandates, or hydrogen blending quotas (e.g., China, Romania)—to ensure stable future demand. As Penelope Goldberg (2025) highlights in the case of electric vehicles, consumer subsidies without local content requirements can create global positive spillovers through learning-by-doing effects. Similarly, for green hydrogen, properly designed demand-side instruments could lower the global cost of electrolyzers and stimulate cross-border technological diffusion. Without this, the world risks repeating the same suboptimal subsidy races that previously limited cooperation in green technology sectors.

A further limitation relates to the lack of standardisation in policy reporting and measurement (Goldberg, 2025). Even when subsidy amounts are disclosed, there is no established framework to determine whether they are “large” or “small.” For example, the high headline figures in U.S. or Chinese subsidies may seem impressive, but once scaled by GDP or population, they appear more modest. Without harmonized metrics, cross-country comparisons are of limited use. Also, since many countries—such as the U.S., Japan, and Australia—deploy multiple overlapping tools, understanding the total effect of policy packages becomes even more complex. Despite these challenges, the clustering model proved valuable in categorising policies and showing patterns that are not immediately visible through raw data alone. The model’s ability to group observations according to their structural similarities—without relying on predefined labels—offers a promising method for future research in green industrial policy. The next step would be to enrich the model with deeper metadata: for example, adding features on policy design, duration, and sector-specific goals, and combining this with supervised learning techniques to assess policy effectiveness over time.

Finally, the results emphasise the importance of international coordination in shaping a successful global green hydrogen market. As with electric vehicles, fragmentation and domestic content requirements could undermine shared progress. Countries—especially developing ones—should be cautious not to replicate the protectionist tendencies seen in past industrial policy waves. Instead, as Goldberg (2025) notes, internationally aligned, innovation-driven policies offer a better chance of creating shared value. For developing countries with strong renewable endowments, export-oriented green hydrogen strategies could provide new industrial paths—if supported by transparent, targeted, and cooperative industrial policy frameworks.

⁸The [Canadian response](#) to the American Inflation Reduction Act production tax credit was not to copy-paste the American approach, but rather launching green hydrogen investment tax credits coupled with contracts-for-differences, a design which the Canadian government deemed to offer a more predictable signal to investors.

5 Conclusion and policy recommendations

The analysis highlights the important role of government support in the green hydrogen industry’s growth, especially during its nascent stages due to a combination of market failures, including environmental externalities and learning-by-doing technology spillovers (e.g., electrolyzers manufacturing). Countries with abundant renewable resources and strategic goals—such as energy security or industrial competitiveness—are increasingly integrating green hydrogen into their policy frameworks. However, empirical evidence shows a significant imbalance between push (supply-side) and pull (demand-side) instruments. While grants and tax incentives dominate policy portfolios, few mechanisms exist to guarantee market uptake, which could limit long-term effectiveness.

The clustering analysis confirms that countries do not differ as much by their identity as by the type of instruments they employ. Yet, several challenges still exist: fragmented policy implementation, unclear production targets, inconsistent data disclosure, and limited coordination at the international level. These problems are particularly pressing for developing countries: They often lack the fiscal space or institutional capacity to implement large-scale GIP, yet may have a dynamic comparative advantage in green hydrogen production due to their renewable endowments.

Policy Recommendations: (1) Rebalance toward demand-side instruments: Countries should complement supply-side subsidies with mechanisms like carbon contracts for difference (CfDs), green hydrogen quotas, or public procurement schemes to create stable demand signals for investors. (2) Improve policy transparency and standardisation: Governments and multilateral institutions should work toward common reporting standards on subsidy types, cost estimates, and sectoral targets to improve comparability and policy learning. (3) Support green hydrogen in developing countries: International organisations should help lower-income countries design export-oriented green hydrogen strategies, with concessional financing, technology transfer, and infrastructure support. (4) Avoid protectionist distortions: Policies that impose domestic content requirements can limit positive international spillovers for all users. Coordination through WTO-compatible frameworks should be prepared to prevent subsidy races and inefficient duplication (to date, the WTO does not have a binding framework regarding industrial policy implementations (Goldberg, 2025)). (5) Invest in better data and policy evaluation tools: Future work should integrate cutting-edge machine learning techniques and econometric models to predict policy effectiveness, using more granular data on project uptake and emissions reductions.

In conclusion, green industrial policy for green hydrogen is not only economically justifiable—it is necessary for the sector to unlock its full potential. But for it to succeed, governments must move *beyond* fragmented, uncoordinated interventions *toward* pragmatic, non-dogmatic, transparent, and globally compatible frameworks that match the scale and the urgency of the climate challenge.

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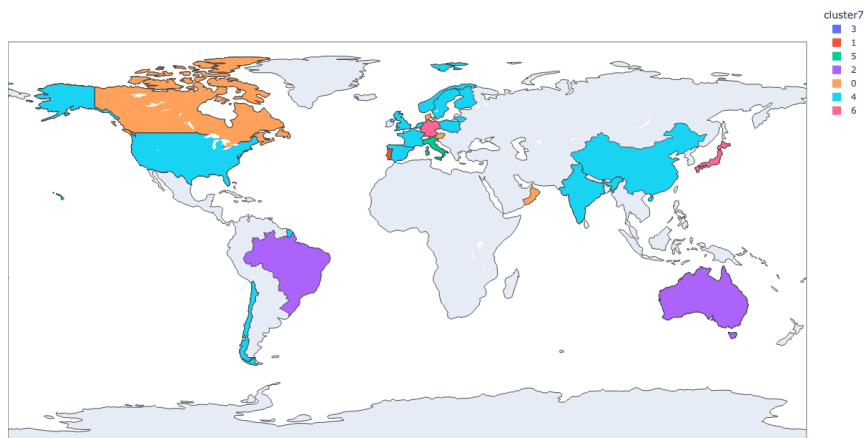
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A Annex: complementary graphs

Hydrogen Policy Clusters by Country (Based on results from the clustering model)



Hydrogen Policy Clusters by Country (Based on results from the clustering model)

